Exact turning invariants for projectile motion under quadratic drag and a vertical-axis Magnus force

Pragyaan Gaur[a)]

Independent researcher

(Dated: 17 August 2026)

A ball spinning about a vertical axis is steered by the Magnus force in the horizontal plane. We show that for constant lift and drag coefficients this problem is exactly integrable in its horizontal part: writing the horizontal velocity as a complex number $w = v_x + iv_y$ and using path length s as the independent variable, the equation of motion becomes the linear, constant-coefficient equation $dw/ds = (ik_L - k_D)w$, in which gravity does not appear. Three exact consequences follow. The heading turns at a constant rate per unit path length, $d\psi/ds = k_L$, so a 180° turn always costs the same path length πL_L with $L_L = 2m/\rho C_L A$, independently of launch speed, launch angle *and* drag. The horizontal speed obeys $|w| = |w_0| \exp[-(C_D/C_L)\Delta\psi]$ exactly, making the horizontal hodograph a logarithmic spiral. The ground-track radius of curvature is $R_h = L_L \cos\gamma$ with γ the flight-path angle, so the track is an oval whose curvature extremum at apex equals L_L. The full system further collapses to a single scalar ordinary differential equation in the turn angle. We apply this to a concrete question: can two balls, released simultaneously in opposite horizontal directions with opposite vertical spin, each turn through 180° and collide head-on at the release height? Without drag the answer is yes, on a one-parameter family we give in closed form. With drag we show the launch-to-meeting chord tips below perpendicular by $(8/3 - 24/\pi^2)\,\mu\theta^2 + O(\mu^2, \theta^4)$, with $\mu = C_D/C_L$; the sign is positive because $\pi^2 > 9$, and the rendezvous therefore fails. Numerically the deficit is strictly positive across $10^{-2} \le \mu \le 3$ and $2^\circ \le \theta \le 88^\circ$. Two oppositely thrown balls thus miss by 1–46 m for real sports balls. All claims are verified against machine-precision numerics; code and data are openly available.

[a)]Electronic mail: pragyaangaur12@gmail.com

I. INTRODUCTION

The lateral deflection of a spinning ball is among the most familiar demonstrations of fluid dynamics in everyday life, and among the oldest quantitatively studied. Briggs measured the sideways deflection of a baseball spinning about a *vertical* axis in a wind tunnel and established its dependence on spin rate and speed[1]; Mehta's review remains the standard entry point to sports-ball aerodynamics[2]; and modern measurements of the lift and drag coefficients of spinning balls are now routine[3,4].

The equations of motion for a spinning ball under gravity, quadratic drag and the Magnus force are simple to write down and are almost universally integrated numerically. Bray and Kerwin, in the standard treatment of a soccer ball struck with sidespin, state the position plainly: "These equations have no closed form solutions but can be solved numerically using a Runge–Kutta routine"[5]. Nathan's reference study of baseball flight likewise proceeds by fourth-order Runge–Kutta[3], as does the recent pedagogical treatment of Mencke *et al.*[6]. In the projectile-dynamics literature proper, Turkyilmazoglu and Altundag obtain closed forms when *either* the quadratic drag or the Magnus term is negligible, and describe the case in which both act simultaneously as highly nonlinear and admitting only perturbative solutions[7]. Chudinov's asymptotic analysis of the same problem obtains the velocity hodograph only as an approximate implicit formula[8]. Reviews of projectile motion with drag[9,10] trace the analytic tradition back to Johann Bernoulli's 1719 device of parametrising planar motion by the tangent angle rather than by time, which decouples the equations at the cost of leaving the solution implicit.

The purpose of this paper is to point out that one physically natural special case of this system is not merely tractable but exactly integrable in elementary functions. When the spin axis is vertical and C_L, C_D are constant, the horizontal motion separates completely from gravity and can be solved in closed form. The resulting invariants are simple, and to our knowledge have not been recorded.

Section II fixes the model. Section III derives the central identity and the exact reduction of the full nine-dimensional system to a single scalar equation. Section IV collects the invariants. Section V poses and answers a concrete geometric question that puts the machinery to work: whether two counter-spinning balls thrown in opposite directions can be made to collide head-on. Section VI describes the numerical verification, and Section VII discusses scope, an open problem, and classroom use.

II. MODEL

A ball of mass m and radius r, with cross-sectional area $A = \pi r^2$, moves through still air of density ρ under gravity, quadratic drag, and the Magnus force,

$$\frac{d\boldsymbol{v}}{dt} = -g\hat{\boldsymbol{z}} - \frac{\rho C_D A}{2m}\,|\boldsymbol{v}|\,\boldsymbol{v} + \frac{\rho C_L A}{2m}\,|\boldsymbol{v}|\,(\hat{\boldsymbol{\omega}} \times \boldsymbol{v}), \tag{1}$$

where $\hat{\boldsymbol{\omega}}$ is the unit spin vector. Throughout we take the spin axis vertical, $\hat{\boldsymbol{\omega}} = \pm\hat{\boldsymbol{z}}$, which is the pure-sidespin configuration of Refs. 1 and 5. It is convenient to absorb the coefficients into two inverse lengths,

$$k_L = \frac{\rho C_L A}{2m}, \qquad k_D = \frac{\rho C_D A}{2m}, \qquad \mu \equiv \frac{k_D}{k_L} = \frac{C_D}{C_L}, \tag{2}$$

and to name the *lift length*

$$L_L = \frac{1}{k_L} = \frac{2m}{\rho C_L A}. \tag{3}$$

The combination (3) is the exact analogue of the radius of a constant-C_L turn in aircraft performance, where both the centripetal requirement and the available lift scale as v^2 and the speed cancels.

We write ψ for the azimuth of the horizontal velocity, measured as an unwrapped angle rather than through an inverse tangent, and s for path length, so that $ds = |\boldsymbol{v}|\,dt$. The flight-path (climb) angle is γ, with $\tan\gamma = v_z/|w|$ where w is defined below. Sections III and IV assume C_L and C_D constant; Section VI reports what happens when that assumption is relaxed.

III. EXACT REDUCTION

A. The central identity

Combine the two horizontal components of the velocity into a single complex number,

$$w = v_x + iv_y. \tag{4}$$

Three observations make the horizontal problem linear. Gravity is vertical and contributes nothing to w. Drag is antiparallel to $\boldsymbol{v}$ and contributes $-k_D|\boldsymbol{v}|w$. And with $\hat{\boldsymbol{\omega}} = \hat{\boldsymbol{z}}$ the

Magnus term is $\hat{\boldsymbol{\omega}} \times \boldsymbol{v} = (-v_y, v_x, 0)$, which is horizontal and, in complex notation, is exactly iw: a rotation by 90°. Hence

$$\frac{dw}{dt} = \big(ik_L - k_D\big)\, |\boldsymbol{v}|\, w. \tag{5}$$

The right-hand side is proportional to $|\boldsymbol{v}|$, which is precisely what makes the change of variable from time to path length effective. Using $ds = |\boldsymbol{v}|dt$, the speed cancels identically and (5) becomes linear with constant coefficients:

$$\boxed{\frac{dw}{ds} = \big(ik_L - k_D\big)\, w \quad \Longrightarrow \quad w(s) = w_0\, e^{-(k_D - ik_L)s}.} \tag{6}$$

Equation (6) is exact. It holds with gravity fully active, at any launch angle, for any ball, and it is independent of the vertical motion. It is the source of everything that follows.

Two features are worth emphasising. First, the quadratic velocity dependence of *both* aerodynamic forces is essential: it is the common factor $|\boldsymbol{v}|$ that cancels under $ds = |\boldsymbol{v}|dt$. A Magnus force linear in speed, as arises for vortices in superfluids where the Magnus–Lorentz correspondence is standard[11], would not have this property, and the associated turning radius would depend on speed. Second, the reparametrisation is in the spirit of Bernoulli's tangent-angle method[9], but here the angle variable is an exact affine function of path length, which is what upgrades the solution from implicit to explicit.

B. Reduction to one scalar equation

Because $|w(s)|$ is known in closed form, the entire system reduces to a single scalar equation. Take the turn angle $u \equiv \psi$ as independent variable (by (6), $u = k_L s$) and let

$$Q(u) \equiv \tan\gamma(u) = \frac{v_z}{|w|}. \tag{7}$$

Since the Magnus force has no vertical component, $\dot{v}_z = -g - k_D|\boldsymbol{v}|v_z$, and dividing by $\dot{\psi} = k_L|\boldsymbol{v}|$ gives $dv_z/du + \mu v_z = -g/(k_L|\boldsymbol{v}|)$. Multiplying by the integrating factor $e^{\mu u}$ and using $|w| = |w_0|e^{-\mu u}$ together with $|w| = |\boldsymbol{v}|\cos\gamma$ yields

$$\boxed{\frac{dQ}{du} = -\lambda\, \frac{e^{2\mu u}}{\sqrt{1+Q^2}}, \qquad Q(0) = \tan\theta,} \tag{8}$$

with the single dimensionless group

$$\lambda = \frac{g}{k_L\, v_0^2 \cos^2\theta} = \frac{gL_L}{v_0^2\cos^2\theta}, \tag{9}$$

θ the launch elevation and v_0 the launch speed. Equation (8) carries the whole problem: given $Q(u)$, the heading is $\psi = u$, the horizontal speed is $|w_0|e^{-\mu u}$, the total speed is $|w_0|e^{-\mu u}/\cos\gamma$, and the position follows by quadrature. A nine-dimensional nonlinear system has been reduced exactly to one scalar equation in two parameters (λ, μ).

Two functionals of the solution encode the geometry we shall need. Since $dz/du = v_z/(k_L|\boldsymbol{v}|) = \sin\gamma/k_L$, the ball returns to its release height after a turn Ψ if and only if

$$H \equiv \int_0^{\Psi} \sin\gamma(u)\, du = 0. \tag{10}$$

Similarly $d(x+iy)/du = w/(k_L|\boldsymbol{v}|) = \cos\gamma\, e^{iu}/k_L$, so the launch-to-landing chord is

$$\text{chord} = \frac{1}{k_L}\int_0^{\Psi} \cos\gamma(u)\, e^{iu}\, du. \tag{11}$$

Equation (11) is exact and is the key formula of Section V.

IV. INVARIANTS

A. Constant turning per unit path length

Taking the argument of (6),

$$\frac{d\psi}{ds} = k_L = \text{const}, \tag{12}$$

so the heading advances by a fixed angle per unit distance travelled. Note what is absent: gravity, launch angle, launch speed, and C_D. Drag is antiparallel to $\boldsymbol{v}$ and therefore contributes nothing to $v_x a_y - v_y a_x$; it slows the ball along the path but cannot bend it. An immediate corollary is that a heading change of $\Delta\psi$ always costs

$$S = \frac{\Delta\psi}{k_L} = \Delta\psi\, L_L, \tag{13}$$

so in particular a 180° turn costs exactly πL_L of path, whatever the launch conditions and whatever the drag. For a baseball with $C_L = 0.20$ this is $\pi L_L = 883.6$ m.

B. Logarithmic-spiral hodograph

Taking the modulus of (6) and eliminating s using (12),

$$|w| = |w_0| \exp\bigl[-\mu\, \Delta\psi\bigr], \qquad \mu = \frac{C_D}{C_L}, \tag{14}$$

so the horizontal velocity traces a logarithmic spiral whose pitch is set entirely by the lift-to-drag ratio. At $\Delta\psi = \pi$ this gives the compact statement $|w_f|/|w_0| = e^{-\pi C_D/C_L}$. We stress that (14) governs the *horizontal* speed. The total speed obeys no such law, because gravity re-accelerates the ball on descent and terminal velocity places a floor under v_z; using $e^{-\pi C_D/C_L}$ for $|\boldsymbol{v}_f|/|\boldsymbol{v}_0|$ underestimates the true ratio by more than an order of magnitude at small C_L/C_D (Table IV).

Equation (14) inverts to

$$\frac{C_D}{C_L} = \frac{\ln\big(|w_0|/|w_f|\big)}{\Delta\psi}, \tag{15}$$

which suggests a measurement: the lift-to-drag ratio of a real ball can be read off a single tracked trajectory by comparing the decay of its horizontal speed with the swing of its heading. Gravity drops out of (15) identically, so no correction for the vertical motion is required.

C. Curvature of the ground track

The radius of curvature of the horizontal projection is $R_h = (ds_h/dt)/(d\psi/dt) = |w|/(k_L|\boldsymbol{v}|)$, and $|w|/|\boldsymbol{v}| = \cos\gamma$, so

$$R_h = L_L \cos\gamma. \tag{16}$$

The track is therefore tightest where the ball is climbing or diving most steeply and flattest at the apex, where $\gamma = 0$ and $R_h = L_L$ exactly. This gives a second measurement: the radius of curvature of the ground track at the apex is L_L directly, without separate knowledge of C_L, m or A. In the drag-free case γ runs from $+\theta$ to $-\theta$, so

$$\frac{R_{\max}}{R_{\min}} = \sec\theta, \tag{17}$$

which quantifies exactly how far the track departs from a circle.

D. Gravity is the sole source of non-circularity

Setting $g = 0$ in (8) gives $\gamma \equiv 0$, hence $R_h \equiv L_L$: the ground track is a *perfect circle of radius* L_L, whether or not drag is present. This is a sharper statement than it may appear, since drag can reduce the speed by any factor without altering the geometry. Table I confirms

TABLE I. Zero-gravity ground track, fitted circle radius versus L_L. Drag changes the speed by more than an order of magnitude and leaves the radius unchanged.

ball	drag	fitted radius (m)	L_L (m)	rel. error	speed change (m/s)
baseball	off	281.267705304	281.267705306	5.5×10^{-12}	$50 \to 50.000$
baseball	on	281.267705317	281.267705306	3.8×10^{-11}	$50 \to 3.200$
frisbee	off	8.075877046	8.075877046	5.4×10^{-12}	$50 \to 50.000$
frisbee	on	8.075877047	8.075877046	8.9×10^{-11}	$50 \to 29.619$

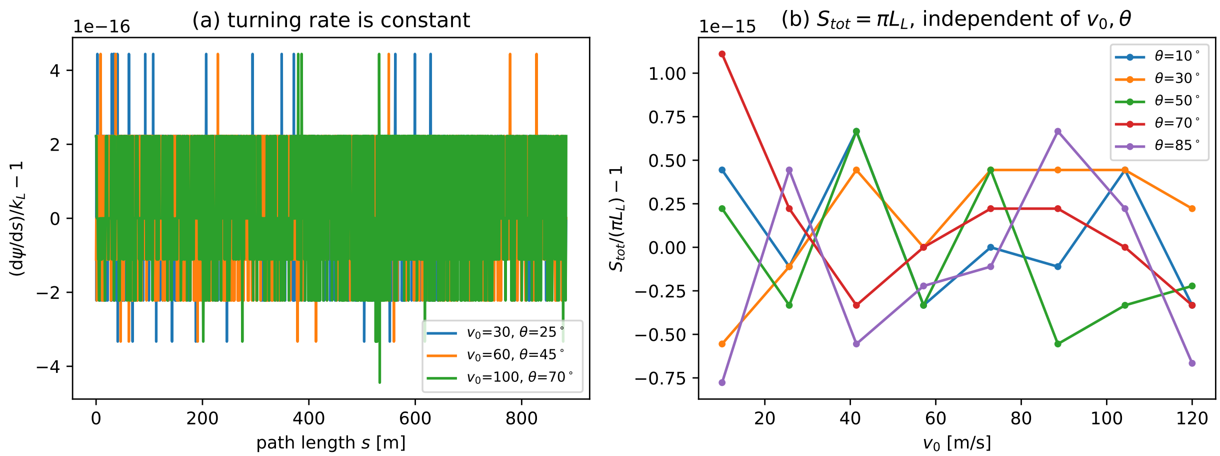

FIG. 1. Numerical verification of the turning invariants, drag off. Left: the turning rate $d\psi/ds$ normalised by k_L, along the path, for three launch conditions; the deviation from unity does not exceed 4.4×10^{-16}. Right: the total path length required for a 180° turn, normalised by πL_L, swept over $v_0 \in [10, 120]$ m/s at five launch angles; the deviation does not exceed 1.1×10^{-15}.

it numerically: a baseball losing 94% of its speed traverses a circle whose fitted radius agrees with L_L to eleven significant figures.

V. THE COUNTER-SPINNING RENDEZVOUS

A. Statement

We now use the machinery on a concrete question. A thrower at the origin releases two identical balls simultaneously in opposite horizontal directions, one spinning about $+\hat{\boldsymbol{z}}$

and the other about $-\hat{\boldsymbol{z}}$. Both curve in the same sense in the horizontal plane. Can launch conditions be chosen so that each ball turns through exactly 180° and the two collide head-on at the height of release?

Label the balls A (launched along bearing 0, spin $+\hat{\boldsymbol{z}}$) and B (bearing π, spin $-\hat{\boldsymbol{z}}$). The configuration is invariant under reflection in the vertical plane at bearing 90° combined with the exchange $A \leftrightarrow B$, since a mirror reflection reverses the horizontal velocity and negates the axial spin vector. Ball B's endpoint is therefore ball A's endpoint reflected in x, and the two coincide if and only if

$$\mathrm{Re}\,(\text{chord}) = 0, \tag{18}$$

that is, if and only if the launch-to-meeting chord is perpendicular to the launch direction. By (11) with $\Psi = \pi$, condition (18) reads

$$\mathcal{I} \equiv \int_0^{\pi} \cos\gamma(u)\, \cos u \, du = 0. \tag{19}$$

The whole question is the sign of a single integral of a positive weight $\cos\gamma$ against $\cos u$; the rendezvous succeeds precisely when that weight is balanced about $u = \pi/2$.

B. Without drag: an exact one-parameter family

For $\mu = 0$ equation (8) is separable, and integrating $\sqrt{1+Q^2}\, dQ = -\lambda\, du$ gives

$$\tfrac{1}{2}\big[Q\sqrt{1+Q^2} + \operatorname{arcsinh} Q\big] = \text{const} - \lambda u. \tag{20}$$

The map $Q(u) \mapsto -Q(\pi - u)$ preserves the $\mu = 0$ equation, so if the solution satisfies $Q(\pi) = -\tan\theta$ then by uniqueness

$$Q(u) = -\,Q(\pi - u) \tag{21}$$

identically. Two things follow at once. First, $\sin\gamma$ is odd about $u = \pi/2$, so the closure condition (10) holds automatically. Second, $\cos\gamma$ is *even* about $u = \pi/2$ while $\cos u$ is odd, so $\mathcal{I} = 0$ exactly: **without drag the chord is always exactly perpendicular to the launch direction, at every launch angle.** The rendezvous is therefore possible, and imposing $Q(\pi) = -\tan\theta$ fixes

$$\lambda = \frac{\tan\theta\sec\theta + \operatorname{arcsinh}(\tan\theta)}{\pi} = \frac{f(\theta)}{\pi\cos^2\theta}, \tag{22}$$

with $f(\theta) = \sin\theta + \cos^2\theta\,\ln(\tan\theta + \sec\theta)$. Combining (22) with (9) gives the launch speed in closed form,

$$v_0 = \sqrt{\frac{\pi g L_L}{f(\theta)}}, \tag{23}$$

for *any* θ: a one-parameter family of solutions. Note that $f(\theta)v_0^2/g$ is the classical arc length of a drag-free parabola, so (23) simply restates (13). For a baseball, (23) ranges from 130.87 m/s at $\theta = 15°$ through a minimum of 85.15 m/s at $\theta = 60°$; the meeting point lies 515.4 m away at $\theta = 45°$.

A detail worth recording, because it is easy to get wrong: at the meeting point both balls are *descending*, so their vertical velocities are common-mode and cancel in the relative velocity. The closing speed is not $2|\boldsymbol{v}_f|$ but

$$|\boldsymbol{v}_A - \boldsymbol{v}_B| = 2|w| = 2\cos\theta\,|\boldsymbol{v}_f|, \tag{24}$$

which we confirm numerically to 6.5×10^{-11} relative over $\theta \in [10°, 80°]$. At $\theta = 60°$ the closing speed equals a single ball's speed.

C. With drag: the chord tips below perpendicular

Once $\mu > 0$ the antisymmetry (21) is broken and $\mathcal{I}$ need not vanish. Because the closure condition (10) determines λ once (μ, θ) are given, the chord bearing at closure is a function of (μ, θ) alone. It cannot depend on C_L and C_D separately, nor on the mass, radius or air density. This is a theorem, not an observation, and it is confirmed numerically to six decimal places for coefficient pairs sharing a common μ.

We evaluate the leading behaviour for small drag and small launch angle. Writing β for the chord bearing, a perturbative solution of (8) (see Appendix A) gives

$$\boxed{90° - \beta \;=\; \left(\frac{8}{3} - \frac{24}{\pi^2}\right)\mu\,\theta^2 \;+\; O(\mu^2, \theta^4)} \tag{25}$$

in radians. The coefficient is $8/3 - 24/\pi^2 = 0.2349583$, and Richardson extrapolation of the numerically computed deficit reproduces it to a relative accuracy of 7×10^{-6} (Fig. 2, right panel).

The sign is the point. The coefficient in (25) is positive if and only if $8/3 > 24/\pi^2$, that is if and only if

$$\pi^2 > 9. \tag{26}$$

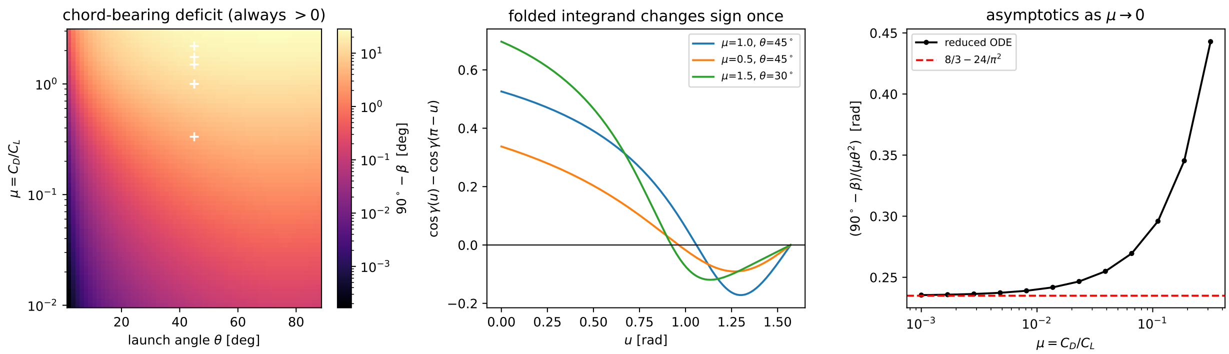

FIG. 2. The obstruction to the rendezvous, computed from the reduced equation (8). Left: the chord-bearing deficit $90° - \beta$ over (μ, θ), strictly positive everywhere, with real sports balls marked at $\theta = 45°$. Middle: the folded integrand of (27), which changes sign once, showing that the positivity of $\mathcal{I}$ is not pointwise. Right: convergence of the numerically computed deficit to the analytic coefficient $8/3 - 24/\pi^2$ of (25) as $\mu \to 0$.

Since $\pi^2 = 9.8696\ldots$, the deficit is positive, the chord tips below perpendicular, and the rendezvous fails. It is a pleasant accident that the obstruction reduces to so elementary an inequality.

D. A tempting argument that does not work

It is natural to try to prove $\mathcal{I} > 0$ pointwise. Folding (19) about $u = \pi/2$,

$$\mathcal{I} = \int_0^{\pi/2} \left[\cos\gamma(u) - \cos\gamma(\pi - u)\right] \cos u \, du, \tag{27}$$

and one might argue that with drag the descent is always steeper than the ascent at matched turn angle, making the bracket positive everywhere. This is false. Because drag makes the ascending arc longer than the descending arc, the apex occurs at a turn angle *greater* than $\pi/2$ (we measure $u_a = 0.543\pi$, 0.572π and 0.635π for the three cases in Fig. 2). For u slightly below $\pi/2$ both u and $\pi - u$ then lie on the ascent, where $|\gamma|$ is decreasing, so the ordering reverses. The bracket in (27) changes sign exactly once (Fig. 2, middle panel). The positivity of $\mathcal{I}$ is genuinely an integral statement, in which a large positive contribution near launch outweighs a smaller negative one near the midpoint. We record this explicitly because the pointwise argument is superficially convincing.

E. Numerical survey and consequences

Beyond the asymptotic regime we evaluate the deficit directly from (8) over a 60×60 grid spanning $10^{-2} \leq \mu \leq 3$ and $2^\circ \leq \theta \leq 88^\circ$, which brackets every ball in Table III. The deficit is strictly positive throughout, with minimum 1.67×10^{-4} degrees at the corner $\mu = 10^{-2}$, $\theta = 2^\circ$, where both small parameters vanish (Fig. 2, left panel). We therefore state:

Result. For constant C_L, C_D with $C_D > 0$ and a vertical spin axis, two balls released in exactly opposite horizontal directions with opposite spin do not collide, at any launch speed or launch angle. This is proved for small $\mu\theta^2$ by (25) and established numerically over the physically relevant domain; a proof for all (μ, θ) remains open.

The practical size of the failure is not small. The two balls miss by $2|\text{chord}|\cos\beta$, which for the closure solutions of Table III is 1.0–1.9 m for a frisbee, 7.6–9.4 m for a table tennis ball and 30–46 m for a soccer ball.

We note in passing that this changes the count of conditions. The two residuals "return to launch height" and "$\psi = \pi$" are two ways of writing the single scalar statement that two events coincide, over the three unknowns (v_0, θ, t^*); the corresponding 2×2 Jacobian is numerically rank one, with singular-value ratio 4.0×10^{-9}. The physical problem nevertheless becomes genuinely two-dimensional once drag is present, through the independent condition (18), which without drag is satisfied automatically.

VI. NUMERICS AND VALIDATION

The full system is integrated as an eleven-component state $[\boldsymbol{r}, \boldsymbol{v}, \boldsymbol{\omega}, \psi, s]$, carrying ψ and s as state variables so that the heading unwraps correctly past $\pm\pi$ without any inverse-tangent differencing. Integration uses explicit Runge–Kutta 4(5) with relative tolerance 10^{-10} and event detection for the two closure events, never fixed steps[12,13]. Parameter sweeps use a batched fixed-step integrator. Figures are produced with Matplotlib[14].

Table II summarises the verification. The reduced equation (8) reproduces $\gamma(\psi)$ from the full three-dimensional integration to better than 10^{-8} rad across four balls and three launch angles, and the reduced chord bearing agrees with the full simulation to better than 10^{-4} degrees. The invariants (12) and (13) hold to machine precision. Crucially, (12) holds equally with drag active, confirming that drag cannot torque the azimuth.

TABLE II. Verification of the analytic results against direct integration. Deviations are relative unless stated.

quantity	condition	max deviation
$d\psi/ds = k_L$	drag off	4.4×10^{-16}
$d\psi/ds = k_L$	drag on	6.7×10^{-16}
$S_{\text{tot}} = \pi L_L$	drag off, $v_0 \times \theta$ sweep	1.1×10^{-15}
$\lvert w\rvert = \lvert w_0\rvert e^{-\mu\Delta\psi}$	drag on, four balls	$< 10^{-9}$
$R_h = L_L \cos\gamma$	drag on and off	$< 10^{-12}$
$g = 0$ track is a circle of radius L_L	drag on and off	$< 10^{-9}$
reduced ODE vs. full 3-D $\gamma(\psi)$	drag on, four balls	$< 10^{-8}$ rad
λ from (22)	drag off	3.5×10^{-12}
v_0 from (23)	drag off	2×10^{-14}
closing speed $= 2\cos\theta\lvert\boldsymbol{v}_f\rvert$	drag off	6.5×10^{-11}
$R_{\text{max}}/R_{\text{min}} = \sec\theta$	drag off	$< 10^{-6}$

A. Beyond constant coefficients

The results of Sections III and IV require C_L and C_D constant. Two remarks bound their scope. First, under constant C_L the Magnus term depends on the spin *direction* only, so the spin magnitude is exactly inert: trajectories for $\omega = 50$, 200 and 800 rad/s coincide to twelve digits. Any question about sensitivity to spin rate is therefore empty until C_L is allowed to depend on the spin parameter $S = r\omega/|\boldsymbol{v}|$. With the saturating form $C_L = 1/(2 + 1/S)$, a 10% spin mismatch between the two balls produces a 1084 m miss at the drag-free 45° solution, larger than the chord itself. Second, we have implemented a smoothed drag-crisis model in which C_D falls from 0.47 to 0.15 across Re ~ 2–4×10^5; with C_D varying, $|w|$ no longer follows (14) exactly, though (12) survives so long as C_L is constant, since drag never contributes to the turning rate whatever its magnitude.

TABLE III. Closure of a single 180° turn with drag active, v_0 capped at 150 m/s; the minimum-v_0 member of the θ-family is shown. Coefficients are representative constants, not measurements.

object	C_L/C_D	closes	θ	v_0 (m/s)	t (s)	v_f/v_0
frisbee	3.00	yes	64°	19.85	2.813	0.590
soccer	1.00	yes	76°	124.20	9.583	0.211
table tennis	0.667	yes	80°	90.95	3.966	0.096
golf	1.00	no	—	—	—	$\psi_{\max} = 0.804\pi$
baseball	0.571	no	—	—	—	$\psi_{\max} = 0.504\pi$
tennis	0.455	no	—	—	—	$\psi_{\max} = 0.565\pi$

TABLE IV. Residual speed at closure: the exact horizontal law (14) compared with the same expression applied, incorrectly, to the total speed.

C_L/C_D	$e^{-\pi C_D/C_L}$	simulated $\lvert\boldsymbol{v}_f\rvert/\lvert\boldsymbol{v}_0\rvert$
0.5	0.0019	0.0435
1.0	0.0432	0.2110
2.0	0.2079	0.4622
3.0	0.3509	0.5938
5.0	0.5335	0.7263
10.0	0.7304	0.8506

B. Real balls

Table III collects the closure solutions with drag active, using representative coefficients[2–4]. Only three of six objects can complete a 180° turn at all below 150 m/s: the required path πL_L must be covered before drag exhausts the ball. A baseball manages 0.504π of turn at best, falling short by 0.496π; a solution exists mathematically only near $v_0 = 616$ m/s, which is Mach 1.8 and outside the validity of an incompressible quadratic-drag model. We emphasise, in view of Section V, that the entries in Table III are single-ball 180° closures and are *not* rendezvous solutions.

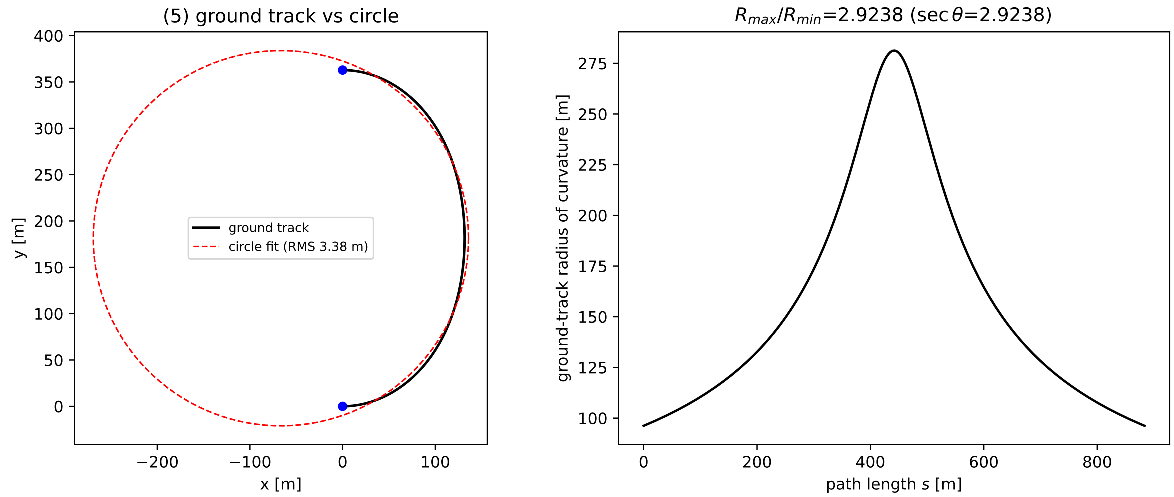

FIG. 3. Ground track of a drag-free closure solution at $\theta = 45°$. Left: the track against its best-fit circle; the RMS residual is 1.585 m on a fitted radius of 264.06 m, which is 43 times the ball radius. Right: the radius of curvature along the path, following $R_h = L_L \cos\gamma$ with $R_{\max}/R_{\min} = \sec\theta$.

VII. DISCUSSION

A. Relation to known results

The individual ingredients here are old. The complex substitution $w = v_x + iv_y$ is standard for rotating-frame problems; the constant-C_L turn radius $2m/\rho C_L A$ is the aircraft-performance result that a wing at fixed lift coefficient turns with a speed-independent radius; and the reparametrisation of projectile motion by a geometric angle is Bernoulli's, dating to 1719[9]. What appears not to have been noticed is that these combine, for the vertical-axis case specifically, into an exactly solvable system: the turning angle is an affine function of path length, so Bernoulli's implicit solution becomes explicit, and the turn radius survives the addition of drag untouched.

It is worth being precise about why the vertical-axis case is special and the general one is not. For a general spin orientation the Magnus term is not a pure rotation of w, the horizontal equation does not close, and (6) fails; this is consistent with the perturbative status of the general problem in Ref. 7. In exterior ballistics the situation is different again: for spin-stabilised projectiles the Magnus *force* is usually small enough to neglect and the relevant lateral deflection arises from the Magnus moment through the yaw of repose[15], a

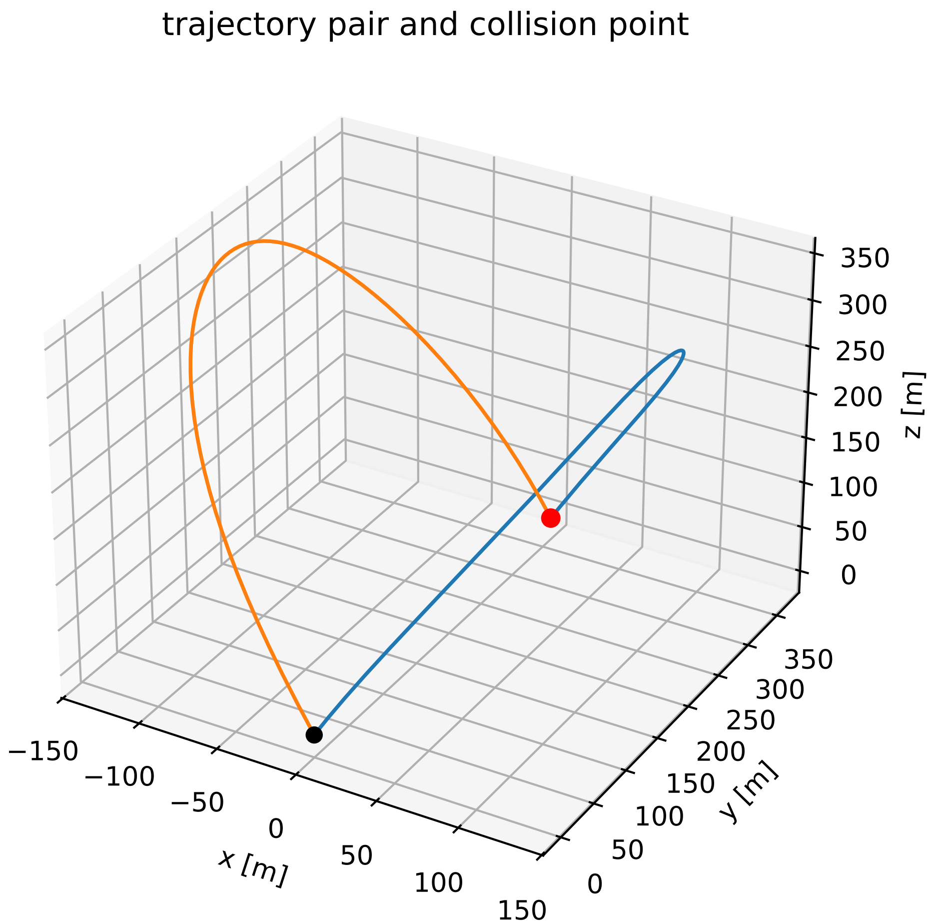

FIG. 4. The drag-free rendezvous: the two counter-spinning trajectories and their common endpoint, at $\theta = 45^\circ$ for a baseball. The mirror symmetry is exact, with arrival times agreeing to 3.6×10^{-15} s.

gyroscopic mechanism unrelated to the one considered here.

The Magnus–Lorentz analogy familiar from vortex dynamics[11] is instructive precisely where it breaks. There the Magnus force is linear in velocity, exactly like the Lorentz force, and the resulting circular orbits have radius proportional to speed. The aerodynamic Magnus force is quadratic, and the radius is L_L regardless of speed. The speed-independence is what makes (13) possible.

B. An open problem

If one drops the requirement that each ball turn through exactly 180° and asks only that two oppositely thrown balls collide head-on at release height, the head-on condition becomes

$$\Delta\psi_A + \Delta\psi_B = 2\pi, \tag{28}$$

so that the *sum* of the turns is fixed while the split need not be equal; the equal split is a drag-free accident. Counting gives four unknowns $(v_{0A}, \theta_A, v_{0B}, \theta_B)$ against four conditions, suggesting isolated solutions with one ball turning past 180°. We searched for these over a 70×70 grid with pairing restricted to $\Delta\psi_A < \pi < \Delta\psi_B$ and found candidates with genuinely unequal turns (145° and 215°), but every refinement drained back to the degenerate corner $\theta \to 0$, $v_0 \to \infty$ and stalled at a residual miss of 1.5–2.0 mm consistent with the θ^2 law of (25). We were unable to establish either existence or non-existence of a finite-angle asymmetric solution, and record it as open.

C. Classroom use

Several of these results are accessible as exercises. Deriving (6) from (1) requires only the observation that $\hat{\boldsymbol{\omega}} \times \boldsymbol{v}$ is a rotation, and makes an unusually clean illustration of why choosing the right independent variable matters. The statement that a 180° turn costs a fixed distance regardless of how hard the ball is thrown is counter-intuitive enough to be worth setting before the derivation. The $g = 0$ circle of Table I is a good computational exercise, as is verifying that drag does not appear in (12). Finally, Section V D is a useful cautionary example of a plausible pointwise argument that fails, with the failure diagnosable by plotting one function.

ACKNOWLEDGMENTS

The author thanks the maintainers of SciPy, NumPy and Matplotlib.

DATA AVAILABILITY

The complete source code, regression test suite and figure-generating scripts are openly available[16]. All numbers quoted in this paper are reproduced by a single driver script, and the analytic claims are encoded as regression tests with the tolerances stated in Table II.

Appendix A: Leading-order deficit

We sketch the calculation leading to (25). Work to first order in μ and second order in θ. For small Q, (8) linearises to $Q' \simeq -\lambda(1+2\mu u)$, so

$$Q(u) \simeq \theta - \lambda\big(u + \mu u^2\big). \tag{A1}$$

Imposing closure (10), $\int_0^\pi Q\,du = 0$ at this order, gives

$$\lambda = \frac{\theta}{\pi/2 + \mu\pi^2/3} \simeq \lambda_0\Big(1 - \frac{2\pi\mu}{3}\Big), \qquad \lambda_0 = \frac{2\theta}{\pi}. \tag{A2}$$

With $\cos\gamma \simeq 1 - Q^2/2$ and $\int_0^\pi \cos u\,du = 0$,

$$\mathcal{I} \simeq -\tfrac{1}{2}\int_0^\pi Q^2 \cos u\,du. \tag{A3}$$

Using $\int_0^\pi u\cos u\,du = -2$, $\int_0^\pi u^2\cos u\,du = -2\pi$ and $\int_0^\pi u^3 \cos u\,du = 12-3\pi^2$, and expanding to first order in μ, the terms independent of μ cancel identically (as they must, by (21)), leaving

$$\int_0^\pi Q^2 \cos u\,du \simeq \mu\theta^2\Big(\frac{16}{3} - 16 + \frac{96}{\pi^2}\Big). \tag{A4}$$

The imaginary part of (11) is $\int_0^\pi \cos\gamma \sin u\,du \simeq 2$ at leading order, so

$$90^\circ - \beta \simeq \frac{\mathcal{I}}{2} = \mu\theta^2\Big(\frac{8}{3} - \frac{24}{\pi^2}\Big), \tag{A5}$$

which is (25). The coefficient is positive because $\pi^2 > 9$.

REFERENCES

[1] L. J. Briggs, "Effect of spin and speed on the lateral deflection (curve) of a baseball; and the Magnus effect for smooth spheres," American Journal of Physics **27**, 589–596 (1959).

[2] R. D. Mehta, "Aerodynamics of sports balls," Annual Review of Fluid Mechanics **17**, 151–189 (1985).

[3]A. M. Nathan, "The effect of spin on the flight of a baseball," American Journal of Physics **76**, 119–124 (2008).

[4]J. R. Kensrud and L. V. Smith, "Drag and lift measurements of solid sports balls in still air," Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology **232**, 236–243 (2018).

[5]K. Bray and D. G. Kerwin, "Modelling the flight of a soccer ball in a direct free kick," Journal of Sports Sciences **21**, 75–85 (2003).

[6]J. E. Mencke, M. Salewski, O. L. Trinhammer, and A. T. Adler, "Flight and bounce of spinning sports balls," American Journal of Physics **88**, 934–947 (2020).

[7]M. Turkyilmazoglu and T. Altundag, "Exact and approximate solutions to projectile motion in air incorporating Magnus effect," The European Physical Journal Plus **135**, 566 (2020).

[8]P. Chudinov, "Study of the asymptotic motion of a sporting projectile taking into account the Magnus force," arXiv preprint **arXiv:2409.15110 [physics.class-ph]** (2024), 10.48550/arXiv.2409.15110.

[9]M. V. Lubarda and V. A. Lubarda, "A review of the analysis of wind-influenced projectile motion in the presence of linear and nonlinear drag force," Archive of Applied Mechanics **92**, 1997–2017 (2022).

[10]P. Chudinov, "On the problem of projectile motion in a medium with quadratic resistance," arXiv preprint **arXiv:2205.01564 [physics.class-ph]** (2022), 10.48550/arXiv.2205.01564.

[11]E. B. Sonin, "Magnus force in superfluids and superconductors," Physical Review B **55**, 485–501 (1997).

[12]P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," Nature Methods **17**, 261–272 (2020).

[13]C. R. Harris *et al.*, "Array programming with NumPy," Nature **585**, 357–362 (2020).

[14]J. D. Hunter, "Matplotlib: A 2D graphics environment," Computing in Science & Engineering **9**, 90–95 (2007).

[15]R. L. McCoy, *Modern Exterior Ballistics: The Launch and Flight Dynamics of Symmetric Projectiles*, 2nd ed. (Schiffer Publishing, Atglen, PA, 2012).

[16]P. Gaur, "magnus-turning-invariants: exact turning invariants for vertical-axis Magnus flight," `https://github.com/` (2026), software and data. DOI to be inserted on deposit.